

Lab 4

1. Address planning
2. Summary/Default routes
3. Troubleshooting static routing

IPv4 address recap

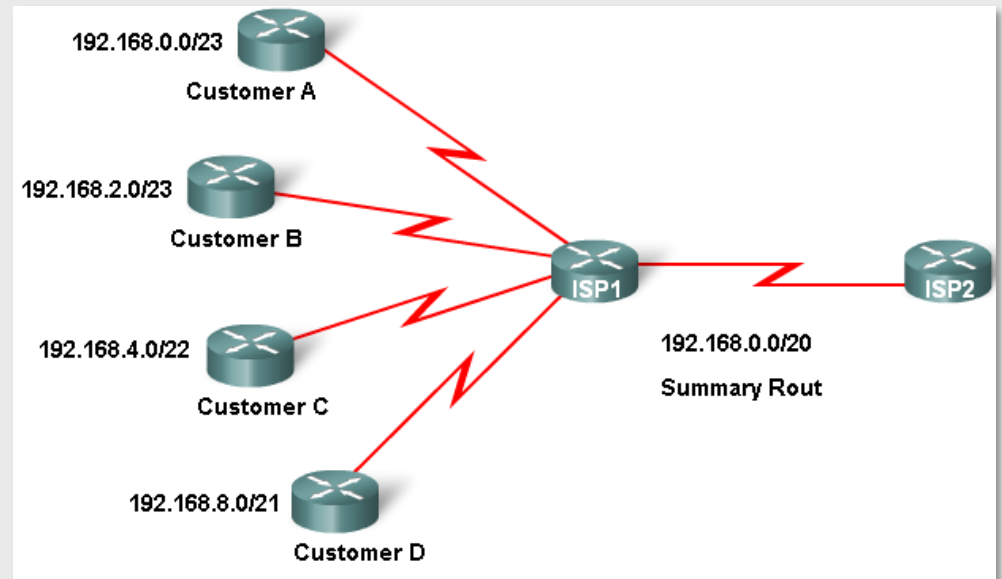
- Classful IPv4 addressing: network address is implicitly determined

IP Address Classes					
Address Class	1st octet range (decimal)	1st octet bits (green bits do not change)	Network(N) and Host(H) parts of address	Default subnet mask (decimal and binary)	Number of possible networks and hosts per network
A	1-127**	00000000-01111111	N.H.H.H	255.0.0.0	128 nets (2^7) 16,777,214 hosts per net (2^{24-2})
B	128-191	10000000-10111111	N.N.H.H	255.255.0.0	16,384 nets (2^{14}) 65,534 hosts per net (2^{16-2})
C	192-223	11000000-11011111	N.N.N.H	255.255.255.0	2,097,150 nets (2^{21}) 254 hosts per net (2^{8-2})
D	224-239	11100000-11101111	NA (multicast)		
E	240-255	11110000-11111111	NA (experimental)		

**** All zeros (0) and all ones (1) are invalid hosts addresses.**

IPv4 address recap

- Classless IPv4 addressing: network address is explicitly determined by a subnet mask
- CIDR: Classless InterDomain Routing [RFC 1517]
 - subnet portion of address of arbitrary length (VLSM: Variable Length Subnet Mask)
 - address format: **a.b.c.d/n**, where **n** is the number of bits in the subnet portion of the address
- More efficient use of IPv4 address space
 - *sub*-subnetting
- Route summarization
 - *supernetting*
 - Prefix aggregation



IPv4 address recap

- Public IP (network) addresses
 - Assigned indirectly by the ISP
 - Assigned directly by IANA (Internet Assigned Numbers Authority)

Class	RFC 1918 Internal Address Range	CIDR Prefix
A	10.0.0.0 - 10.255.255.255	10.0.0.0/8
B	172.16.0.0 - 172.31.255.255	172.16.0.0/12
C	192.168.0.0 - 192.168.255.255	192.168.0.0/16

- IANA set aside three blocks for **private IP addresses** [RFC 1918]
 - Use is unrestricted in private networks
 - Cannot be routed on the public Internet (ISP routers are configured accordingly)
 - Global uniqueness is not guaranteed
- Advantages and disadvantages of using private address space
 - Flexibility in network design due to larger address space at disposal
 - Safe choice once outside connectivity is needed
 - but Network Address Translation is needed then
 - May require renumbering in case of several networks merging

IPv4 address recap

■ Special addresses [RFC 3330]

■ Network and Broadcast addresses

- All 0's host address part
- All 1's host address part

■ "This" network

- **0.0.0.0/8**, refer to source hosts on "this" network

■ Loopback

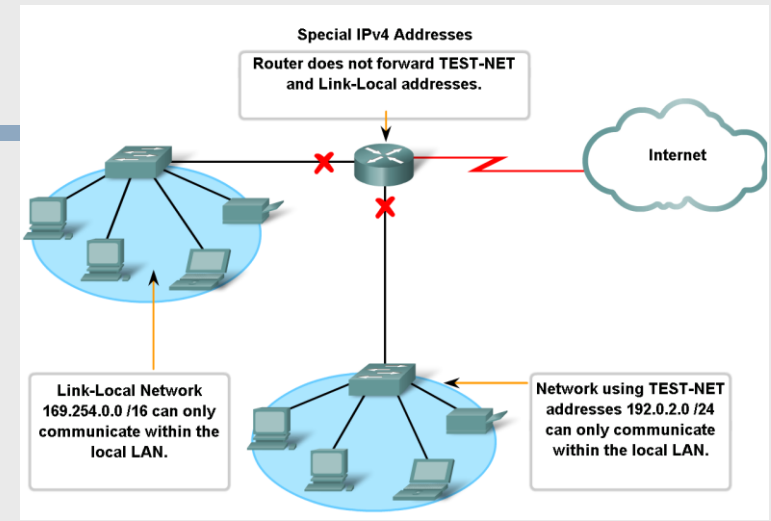
- **127.0.0.1**, is used by hosts to direct traffic to themselves (bypassing lower layers of the stack)
- **127.0.0.0/8** is actually reserved, no address within this block should ever appear on any network

■ Link-local [RFC 3927]

- **169.254.0.0/16**, can be automatically assigned by a local host to itself when no other IP configuration means is available
- Must not be used as destination addresses in packets forwarded to a router, and TTL must be set to 1

■ TEST-NET addresses

- **192.0.2.0/24**, set aside for use in documentation and example code
- Should not appear on the Internet



Subnet addresses calculation

■ 172.16.20.0/25 network

Network address

172 . 16. 20. 0/25

10101100.00010000.00010100.00000000

|-----Network-----| host -|

$0+0+0+0+0+0+0+0=0$

Network address = 172.16.20.0

First host address

172 . 16. 20. 1

10101100.00010000.00010100.00000001

|-----Network-----| host -|

$0+0+0+0+0+0+0+1=1$

Lowest host address = 172.16.20.1

Broadcast address

172 . 16. 20. 127

10101100.00010000.00010100.01111111

|-----Network-----| host -|

$0+64+32+16+8+4+2+1=127$

Broadcast address = 172.16.20.127

Last host address

172 . 16. 20. 126

10101100.00010000.00010100.01111110

|-----Network-----| host -|

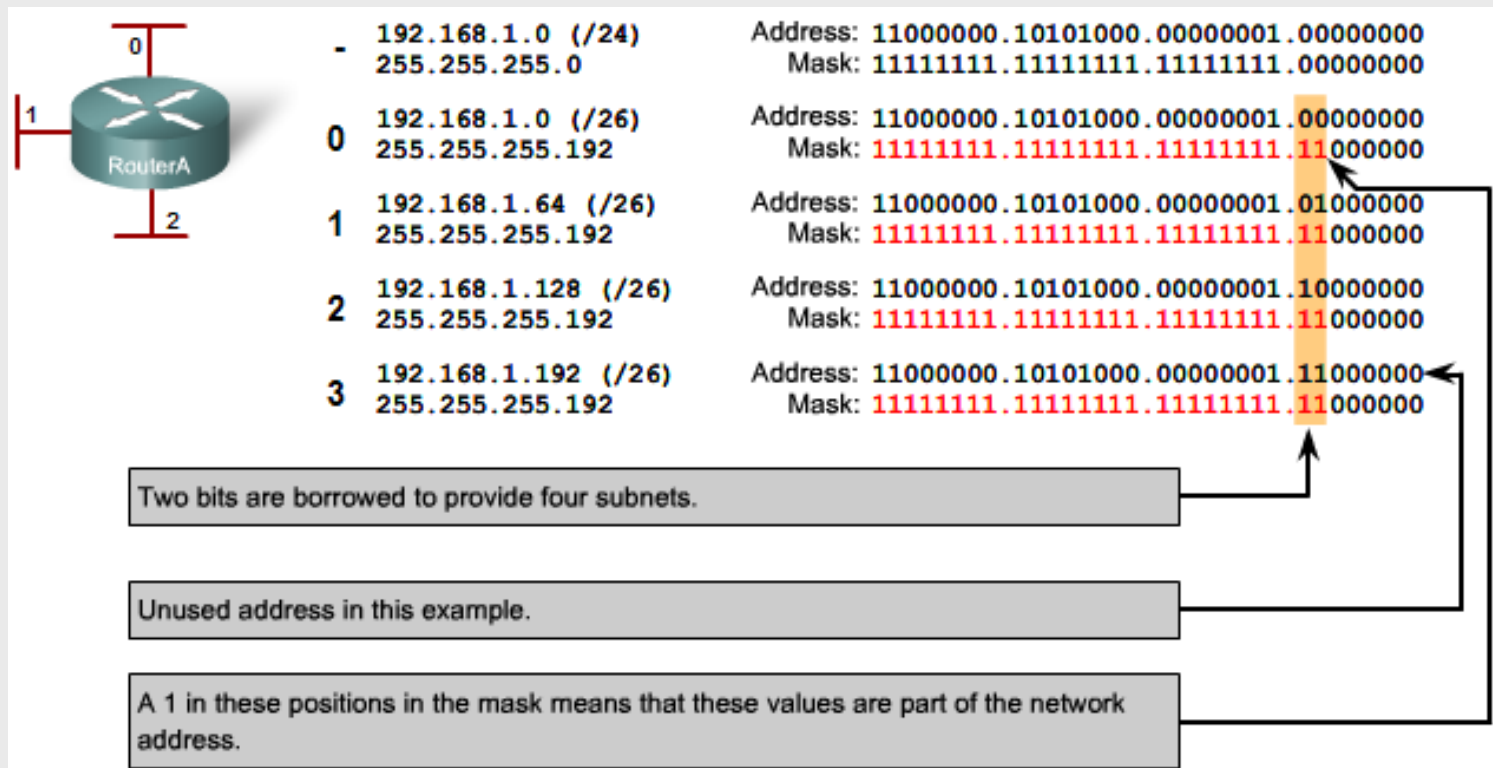
$0+64+32+16+8+4+2+0=126$

Highest host address = 172.16.20.126

Basic subnetting

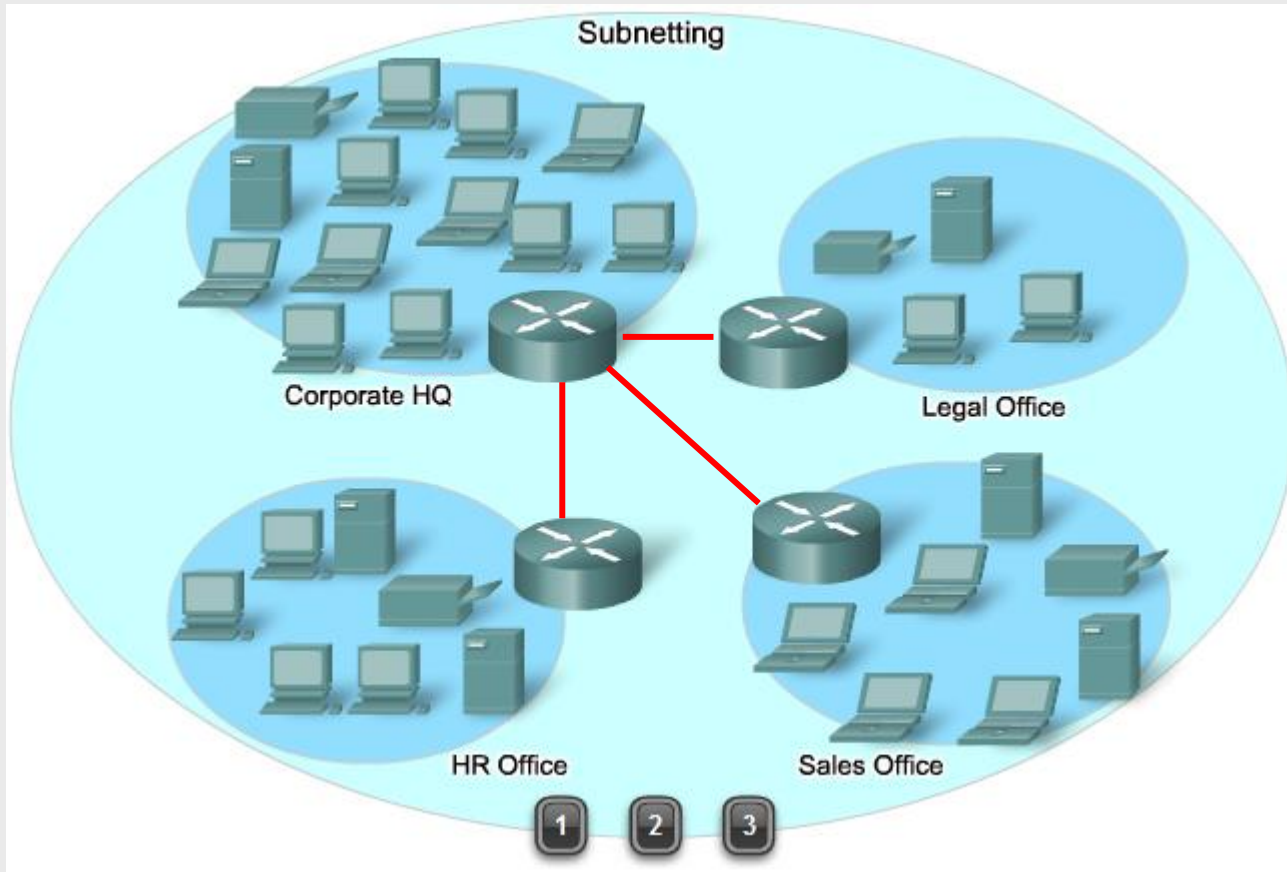
Addressing Scheme: Example of 4 networks

Subnet	Network address	Host range	Broadcast address
0	192.168.1.0/26	192.168.1.1 - 192.168.1.62	192.168.1.63
1	192.168.1.64/26	192.168.1.65 - 192.168.1.126	192.168.1.127
2	192.168.1.128/26	192.168.1.129 - 192.168.1.190	192.168.1.191
3	192.168.1.192/26	192.168.1.193 - 192.168.1.254	192.168.1.255



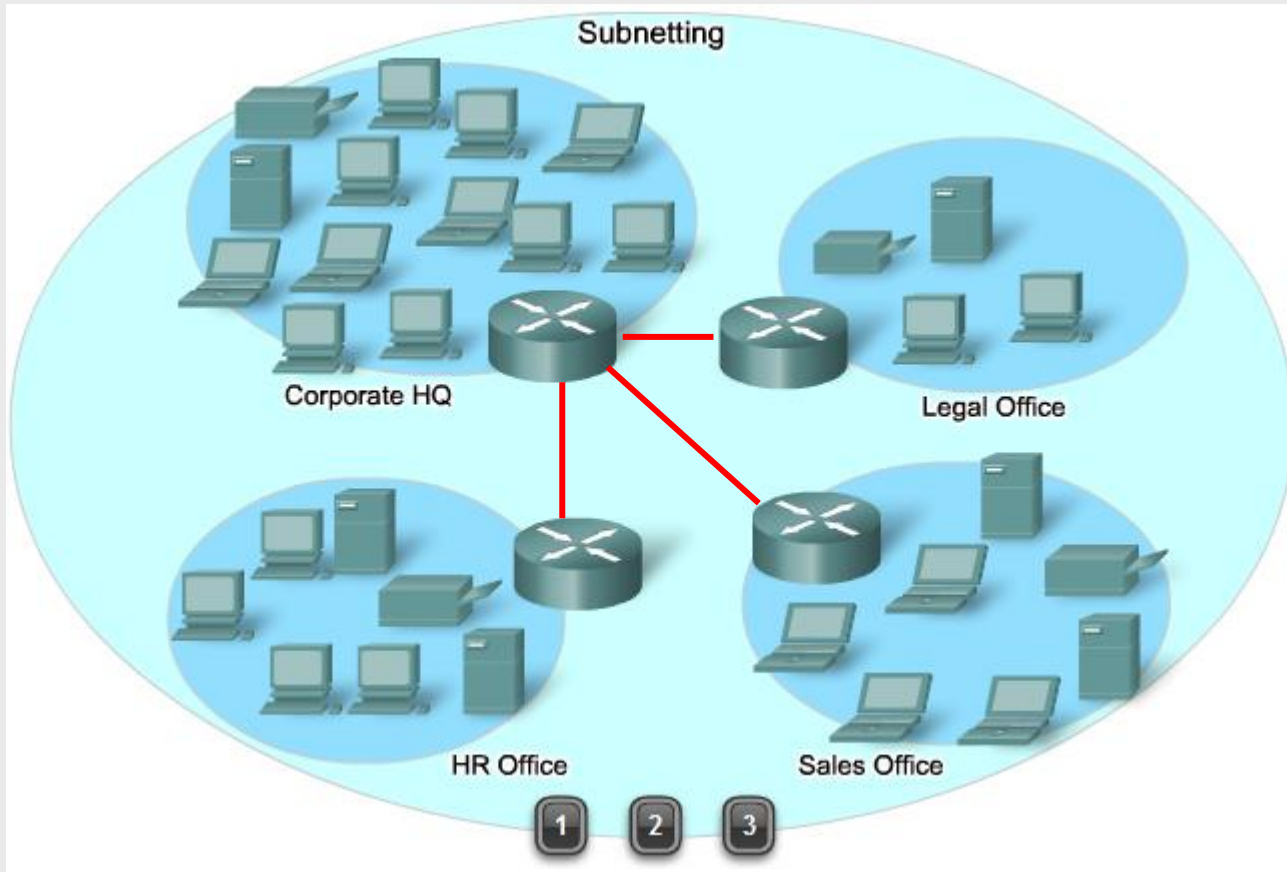
Addressing scheme planning

- Subnetting the available IPv4 address block to efficiently accommodate the required number of hosts of each network



Addressing scheme planning

- Subnetting the available IPv4 address block to efficiently accommodate the required number of hosts of each network

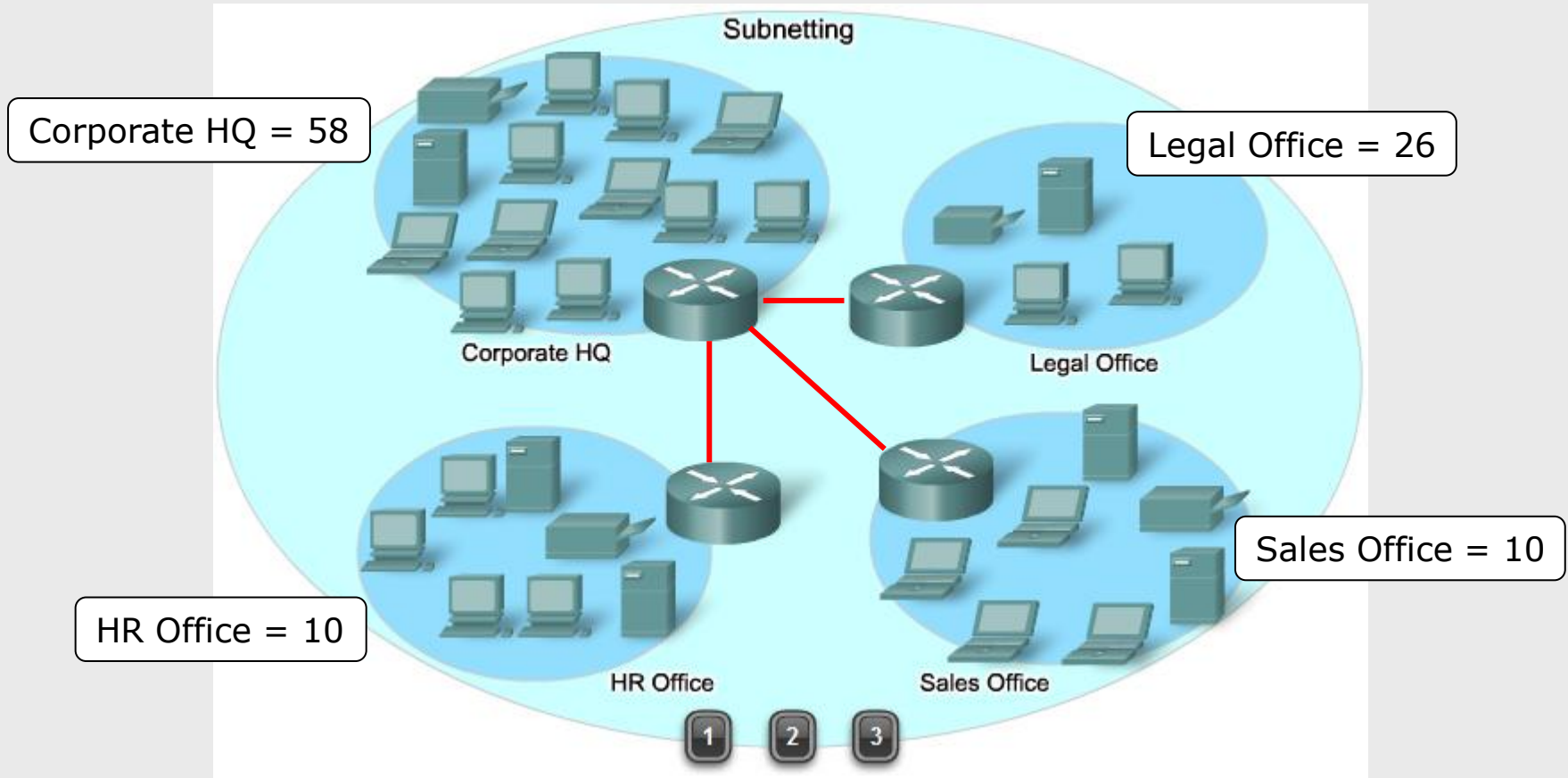


Total number of hosts in the corporate network = 104

Allocate a /?? block

Addressing scheme planning

- Subnetting the available IPv4 address block to efficiently accommodate the required number of hosts of each network

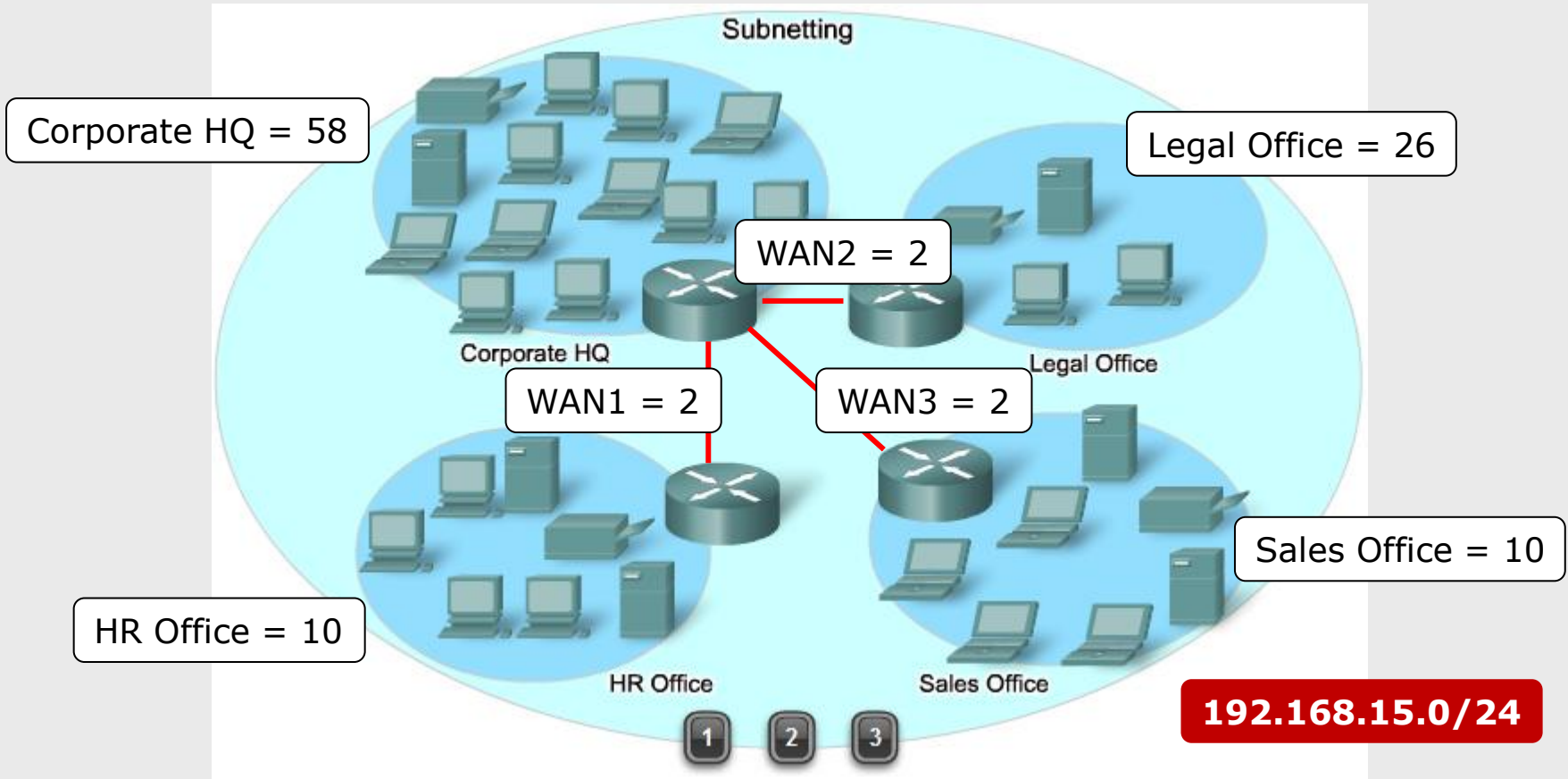


Total number of hosts in the corporate network = 104

Allocate a /?? block

Addressing scheme planning

- Subnetting the available IPv4 address block to efficiently accommodate the required number of hosts of each network



Total number of hosts in the corporate network = 104

Allocate a /24 block

Allocating addresses

- VLSM chart: proceed by decreasing size of the address block

Corporate HQ = 58

	/25 (1 subnet bit) 2 subnets 126 hosts	/26 (2 subnet bits) 4 subnets 62 hosts	/27 (3 subnet bits) 8 subnets 30 hosts	/28 (4 subnet bits) 16 subnets 14 hosts	/29 (5 subnet bits) 32 subnets 6 hosts	/30 (6 subnet bits) 64 subnets 2 hosts
.0	.0	.0 (.1- .62)	.0 (.1- .30)	.0 (.1- .14)	.0 (.1- .6)	.0 (.1- .2)
.4						.4 (.5- .6)
.8					.8 (.9- .14)	.8 (.9- .10)
.12						.12 (.13- .14)
.16				.16 (.17- .22)		.16 (.17- .18)
.20						.20 (.21- .22)
.24					.24 (.25- .26)	
.28					.28 (.29- .30)	
.32			.32 (.33- .62)	.32 (.33- .38)	.32 (.33- .34)	
.36					.36 (.37- .38)	
.40					.40 (.41- .42)	
.44					.44 (.45- .46)	
.48				.48 (.49- .54)	.48 (.49- .50)	
.52					.52 (.53- .54)	
.56					.56 (.57- .58)	
.60					.60 (.61- .62)	
.64		.64 (.65- .126)	.64 (.65- .94)	.64 (.65- .78)	.64 (.65- .70)	.64 (.65- .66)
.68						.68 (.69- .70)
.72					.72 (.73- .78)	.72 (.73- .74)
.76						.76 (.77- .78)
.80				.80 (.81- .86)		.80 (.81- .82)
.84						.84 (.85- .86)
.88					.88 (.89- .94)	.88 (.89- .90)
.92						.92 (.93- .94)
.96			.96 (.97- .102)	.96 (.97- .98)		
.100				.100 (.101- .102)		
.104				.104 (.105- .110)	.104 (.105- .106)	
.108					.108 (.109- .110)	
.112					.112 (.113- .118)	.112 (.113- .114)
.116						.116 (.117- .118)
.120				.120 (.121- .126)		.120 (.121- .122)
.124						.124 (.125- .126)

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ee Toderick

Allocating addresses

- VLSM chart: proceed by decreasing size of the address block

Corporate HQ = 58

192.168.15.0/26

	/25 (1 subnet bit) 2 subnets 126 hosts	/26 (2 subnet bits) 4 subnets 62 hosts	/27 (3 subnet bits) 8 subnets 30 hosts	/28 (4 subnet bits) 16 subnets 14 hosts	/29 (5 subnet bits) 32 subnets 6 hosts	/30 (6 subnet bits) 64 subnets 2 hosts	
.0	.0	.0 (.1- .62)	.0 (.1- .30)	.0 (.1- .14)	.0 (.1- .6)	.0 (.1- .2)	
.4						.4 (.5- .6)	
.8					.16 (.17- .30)	.8 (.9- .14)	.8 (.9- .10)
.12							.12 (.13- .14)
.16				.32 (.33- .62)		.16 (.17- .22)	.16 (.17- .18)
.20							.20 (.21- .22)
.24					.48 (.49- .62)	.24 (.25- .30)	.24 (.25- .26)
.28							.28 (.29- .30)
.32			.32 (.33- .38)	.32 (.33- .34)			
.36				.36 (.37- .38)			
.40			.64 (.65- .94)	.40 (.41- .46)	.40 (.41- .42)		
.44					.44 (.45- .46)		
.48					.80 (.81- .94)	.48 (.49- .50)	
.52						.52 (.53- .54)	
.56				.56 (.57- .62)		.56 (.57- .58)	
.60						.60 (.61- .62)	
.64		.64 (.65- .126)		.64 (.65- .94)	.64 (.65- .78)	.64 (.65- .70)	.64 (.65- .66)
.68							.68 (.69- .70)
.72			.80 (.81- .94)			.72 (.73- .78)	.72 (.73- .74)
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.80					.96 (.97- .126)	.80 (.81- .86)	.80 (.81- .82)
.84							.84 (.85- .86)
.88			.112 (.113- .126)			.88 (.89- .94)	.88 (.89- .90)
.92							.92 (.93- .94)
.96				.120 (.121- .126)	.96 (.97- .102)	.96 (.97- .98)	
.100						.100 (.101- .102)	
.104			.104 (.105- .110)		.104 (.105- .106)		
.108					.108 (.109- .110)		
.112			.112 (.113- .118)	.112 (.113- .118)	.112 (.113- .114)		
.116					.116 (.117- .118)		
.120				.120 (.121- .126)	.120 (.121- .122)		
.124					.124 (.125- .126)		

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Allocating addresses

- VLSM chart: proceed by decreasing size of the address block

Legal Office = 26

	/25 (1 subnet bit) 2 subnets 126 hosts	/26 (2 subnet bits) 4 subnets 62 hosts	/27 (3 subnet bits) 8 subnets 30 hosts	/28 (4 subnet bits) 16 subnets 14 hosts	/29 (5 subnet bits) 32 subnets 6 hosts	/30 (6 subnet bits) 64 subnets 2 hosts
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.8				.16 (.17- .30)	.8 (.9- .14)	.8 (.9- .10)
.12						.12 (.13- .14)
.16			.32 (.33- .62)	.32 (.33- .46)	.16 (.17- .22)	.16 (.17- .18)
.20						.20 (.21- .22)
.24					.24 (.25- .30)	.24 (.25- .26)
.28						.28 (.29- .30)
.32				.48 (.49- .62)	.32 (.33- .38)	.32 (.33- .34)
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.40					.40 (.41- .46)	.40 (.41- .42)
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.48					.48 (.49- .54)	.48 (.49- .50)
.52						.52 (.53- .54)
.56					.56 (.57- .62)	.56 (.57- .58)
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Allocating addresses

- VLSM chart: proceed by decreasing size of the address block

Legal Office = 26

192.168.15.64/27

	/25 (1 subnet bit) 2 subnets 126 hosts	/26 (2 subnet bits) 4 subnets 62 hosts	/27 (3 subnet bits) 8 subnets 30 hosts	/28 (4 subnet bits) 16 subnets 14 hosts	/29 (5 subnet bits) 32 subnets 6 hosts	/30 (6 subnet bits) 64 subnets 2 hosts	
.0	.0	.0 (.1- .62)	.0 (.1- .30)	.0 (.1- .14)	.0 (.1- .6)	.0 (.1- .2)	
.4						.4 (.5- .6)	
.8				.16 (.17- .30)	.8 (.9- .14)	.8 (.9- .10)	
.12						.12 (.13- .14)	
.16			.32 (.33- .62)	.32 (.33- .46)	.16 (.17- .22)	.16 (.17- .18)	
.20					.24 (.25- .30)	.20 (.21- .22)	
.24				.48 (.49- .62)		.24 (.25- .26)	
.28					.32 (.33- .38)	.28 (.29- .30)	
.32					.40 (.41- .46)	.32 (.33- .34)	
.36						.36 (.37- .38)	
.40				.64 (.65- .78)	.48 (.49- .54)	.40 (.41- .42)	
.44					.56 (.57- .62)	.44 (.45- .46)	
.48						.48 (.49- .50)	
.52					.64 (.65- .70)	.52 (.53- .54)	
.56				.80 (.81- .94)	.72 (.73- .78)	.56 (.57- .58)	
.60						.60 (.61- .62)	
.64		.64 (.65- .126)	.64 (.65- .94)		.80 (.81- .86)	.64 (.65- .66)	
.68					.88 (.89- .94)	.68 (.69- .70)	
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.100					.104 (.105- .110)	.100 (.101- .102)	
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.108					.112 (.113- .118)	.108 (.109- .110)	
.112						.112 (.113- .114)	
.116					.120 (.121- .126)	.116 (.117- .118)	
.120						.120 (.121- .122)	
.124						.124 (.125- .126)	

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Allocating addresses

- VLSM chart: proceed by decreasing size of the address block

Sales Office = 10

HR Office = 10

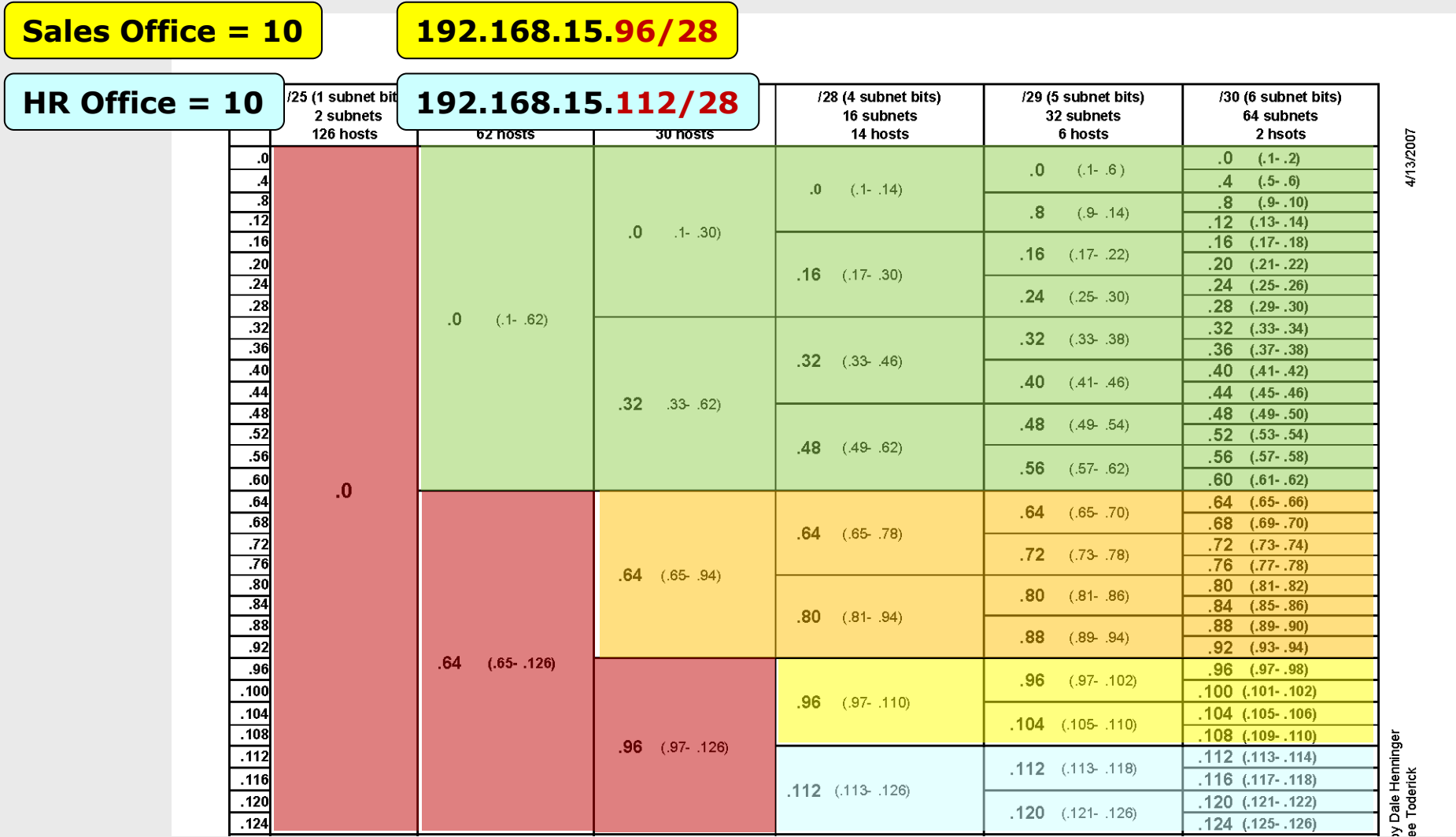
	/25 (1 subnet bit) 2 subnets 126 hosts	/26 (2 subnet bits) 4 subnets 62 hosts	/27 (3 subnet bits) 8 subnets 30 hosts	/28 (4 subnet bits) 16 subnets 14 hosts	/29 (5 subnet bits) 32 subnets 6 hosts	/30 (6 subnet bits) 64 subnets 2 hosts
.0	.0	.0 (.1- .62)	.0 (.1- .30)	.0 (.1- .14)	.0 (.1- .6)	.0 (.1- .2)
.4						.4 (.5- .6)
.8					.8 (.9- .14)	.8 (.9- .10)
.12						.12 (.13- .14)
.16			.16 (.17- .30)	.16 (.17- .30)	.16 (.17- .22)	.16 (.17- .18)
.20						.20 (.21- .22)
.24				.24 (.25- .30)	.24 (.25- .30)	.24 (.25- .26)
.28						.28 (.29- .30)
.32		.32 (.33- .62)	.32 (.33- .46)	.32 (.33- .46)	.32 (.33- .38)	.32 (.33- .34)
.36						.36 (.37- .38)
.40				.40 (.41- .46)	.40 (.41- .46)	.40 (.41- .42)
.44						.44 (.45- .46)
.48			.48 (.49- .62)	.48 (.49- .54)	.48 (.49- .54)	.48 (.49- .50)
.52						.52 (.53- .54)
.56				.56 (.57- .62)	.56 (.57- .62)	.56 (.57- .58)
.60						.60 (.61- .62)
.64	.64	.64 (.65- .126)	.64 (.65- .94)	.64 (.65- .78)	.64 (.65- .70)	.64 (.65- .66)
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.76						.76 (.77- .78)
.80			.80 (.81- .94)	.80 (.81- .86)	.80 (.81- .86)	.80 (.81- .82)
.84						.84 (.85- .86)
.88				.88 (.89- .94)	.88 (.89- .94)	.88 (.89- .90)
.92						.92 (.93- .94)
.96			.96 (.97- .126)	.96 (.97- .110)	.96 (.97- .102)	.96 (.97- .98)
.100						.100 (.101- .102)
.104					.104 (.105- .110)	.104 (.105- .106)
.108						.108 (.109- .110)
.112				.112 (.113- .126)	.112 (.113- .118)	.112 (.113- .114)
.116						.116 (.117- .118)
.120					.120 (.121- .126)	.120 (.121- .122)
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Allocating addresses

- VLSM chart: proceed by decreasing size of the address block



Allocating addresses

- VLSM chart: proceed by decreasing size of the address block

WAN 1= 2

WAN 2= 2

WAN 3= 2

	/25 (1 subnet bit) 2 subnets 126 hosts	/26 (2 subnet bits) 4 subnets 62 hosts	/27 (3 subnet bits) 8 subnets 30 hosts	/28 (4 subnet bits) 16 subnets 14 hosts	/29 (5 subnet bits) 32 subnets 6 hosts	/30 (6 subnet bits) 64 subnets 2 hosts
.112			.112 (.112- .126)	.112 (.113- .126)	.112 (.113- .118)	.112 (.113- .114)
.116						
.120						
.124						
.128	.128	.128 (.129- .190)	.128 (.129- .158)	.128 (.129- .142)	.128 (.129- .134)	.128 (.129- .130)
.132						
.136						
.140						
.144						
.148						
.152						
.156						
.160						
.164						
.168						
.172						
.176						
.180						
.184						
.188						
.192		.192 (.193- .254)	.192 (.193- .222)	.192 (.193- .206)	.192 (.193- .198)	.192 (.193- .194)
.196						
.200						
.204						
.208						
.212						
.216						
.220						

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Allocating addresses

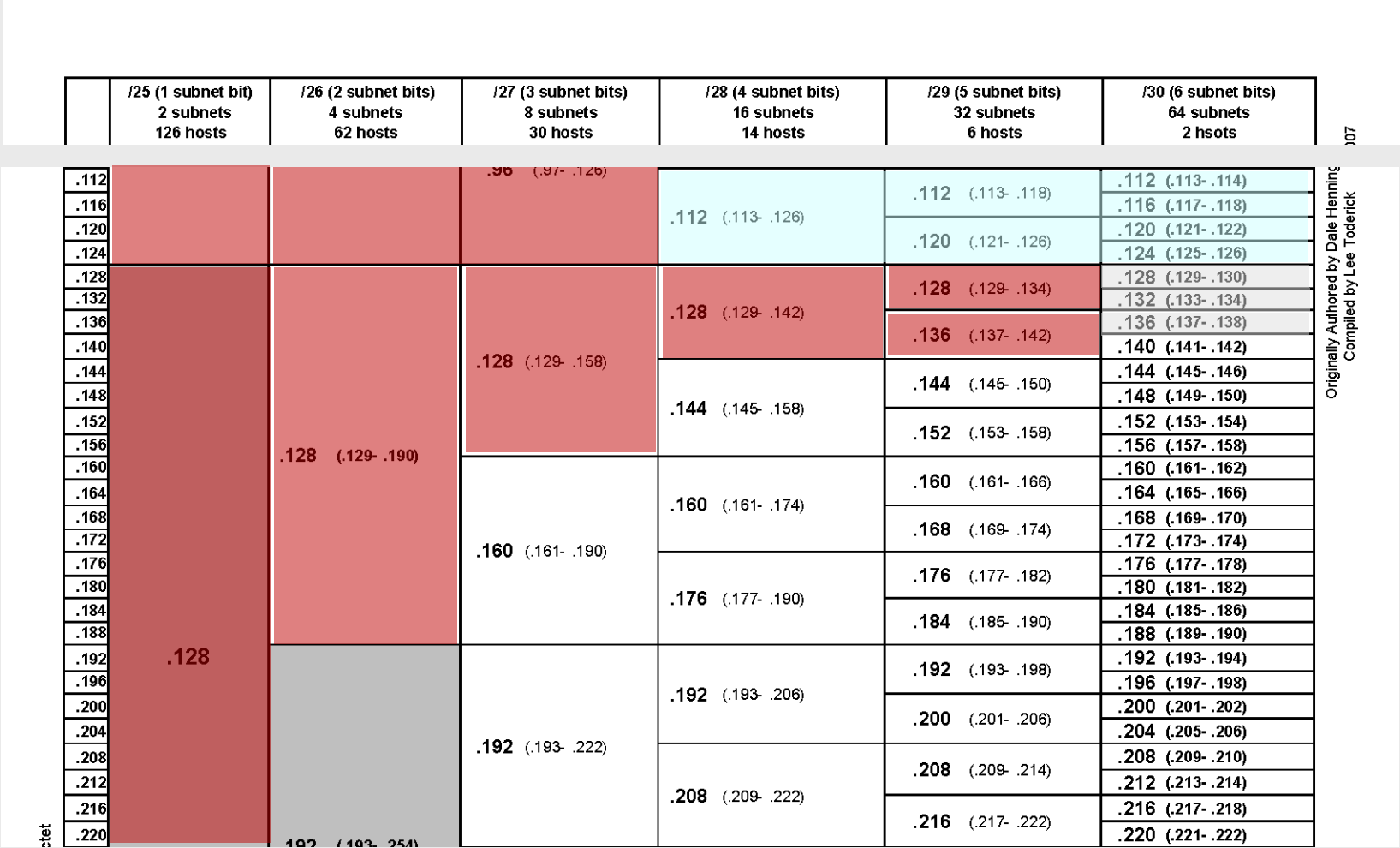
- VLSM chart: proceed by decreasing size of the address block

WAN 1= 2	192.168.15.128/30					
WAN 2= 2	192.168.15.132/30					
WAN 3= 2	192.168.15.136/30					
	/25 (1 subnet bit)	/26 (2 subnet bits)	/27 (3 subnet bits) 8 subnets 30 hosts	/28 (4 subnet bits) 16 subnets 14 hosts	/29 (5 subnet bits) 32 subnets 6 hosts	/30 (6 subnet bits) 64 subnets 2 hosts
.112			.128 (.129-.158)	.112 (.113-.126)	.112 (.113-.118)	.112 (.113-.114)
.116						.116 (.117-.118)
.120					.120 (.121-.126)	.120 (.121-.122)
.124						.124 (.125-.126)
.128				.128 (.129-.142)	.128 (.129-.134)	.128 (.129-.130)
.132					.136 (.137-.142)	.132 (.133-.134)
.136						.136 (.137-.138)
.140			.128 (.129-.158)		.144 (.145-.150)	.140 (.141-.142)
.144				.144 (.145-.158)		.144 (.145-.146)
.148					.152 (.153-.158)	.148 (.149-.150)
.152						.152 (.153-.154)
.156						.156 (.157-.158)
.160		.128 (.129-.190)		.160 (.161-.174)	.160 (.161-.166)	.160 (.161-.162)
.164					.168 (.169-.174)	.164 (.165-.166)
.168						.168 (.169-.170)
.172			.160 (.161-.190)		.176 (.177-.182)	.172 (.173-.174)
.176				.176 (.177-.190)		.176 (.177-.178)
.180					.184 (.185-.190)	.180 (.181-.182)
.184						.184 (.185-.186)
.188						.188 (.189-.190)
.192				.192 (.193-.206)	.192 (.193-.198)	.192 (.193-.194)
.196					.200 (.201-.206)	.196 (.197-.198)
.200						.200 (.201-.202)
.204			.192 (.193-.222)		.208 (.209-.214)	.204 (.205-.206)
.208				.208 (.209-.222)		.208 (.209-.210)
.212					.216 (.217-.222)	.212 (.213-.214)
.216						.216 (.217-.218)
.220						.220 (.221-.222)

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Allocating addresses

- VLSM chart: proceed by decreasing size of the address block



Addressing scheme planning

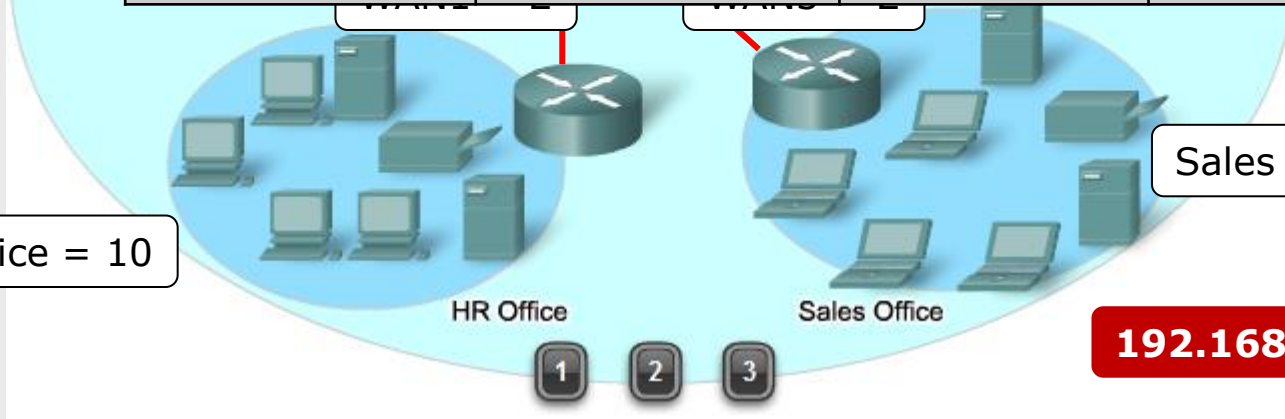
■ Subnetting accommodate

Req. addresses	Network prefix	Address range	Broadcast address
Corporate HQ (58)	192.168.15.0/26	.1-.62	.63
Legal Office (26)	192.168.15.64/27	.65-.94	.95
Sales Office (10)	192.168.15.96/28	.97-.110	.111
HR Office (10)	192.168.15.112/28	.113-.126	.127
WAN1 (2)	192.168.15.128/30	.129-.130	.131
WAN2 (2)	192.168.15.132/30	.133-.134	.135
WAN3 (2)	192.168.15.136/30	.137-.138	.139

Corporate HQ = 58

HR Office = 10

Sales Office = 10

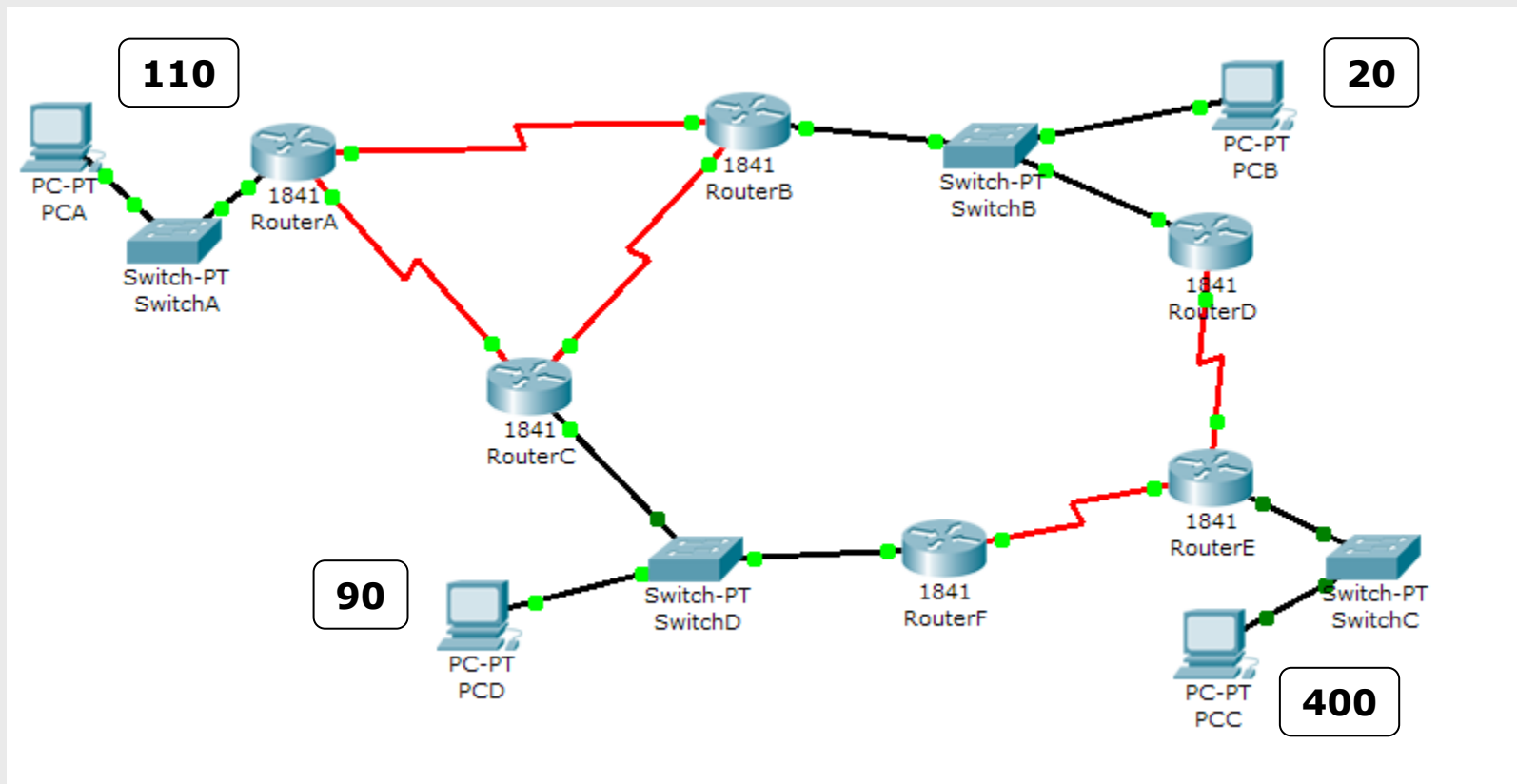


Total number of hosts in the corporate network = 104

Allocate a /24 block

Lab activity

- The block of addresses **172.16.0.0/??** is given
 - Design an addressing scheme
 - *Design and configure static routing accordingly*



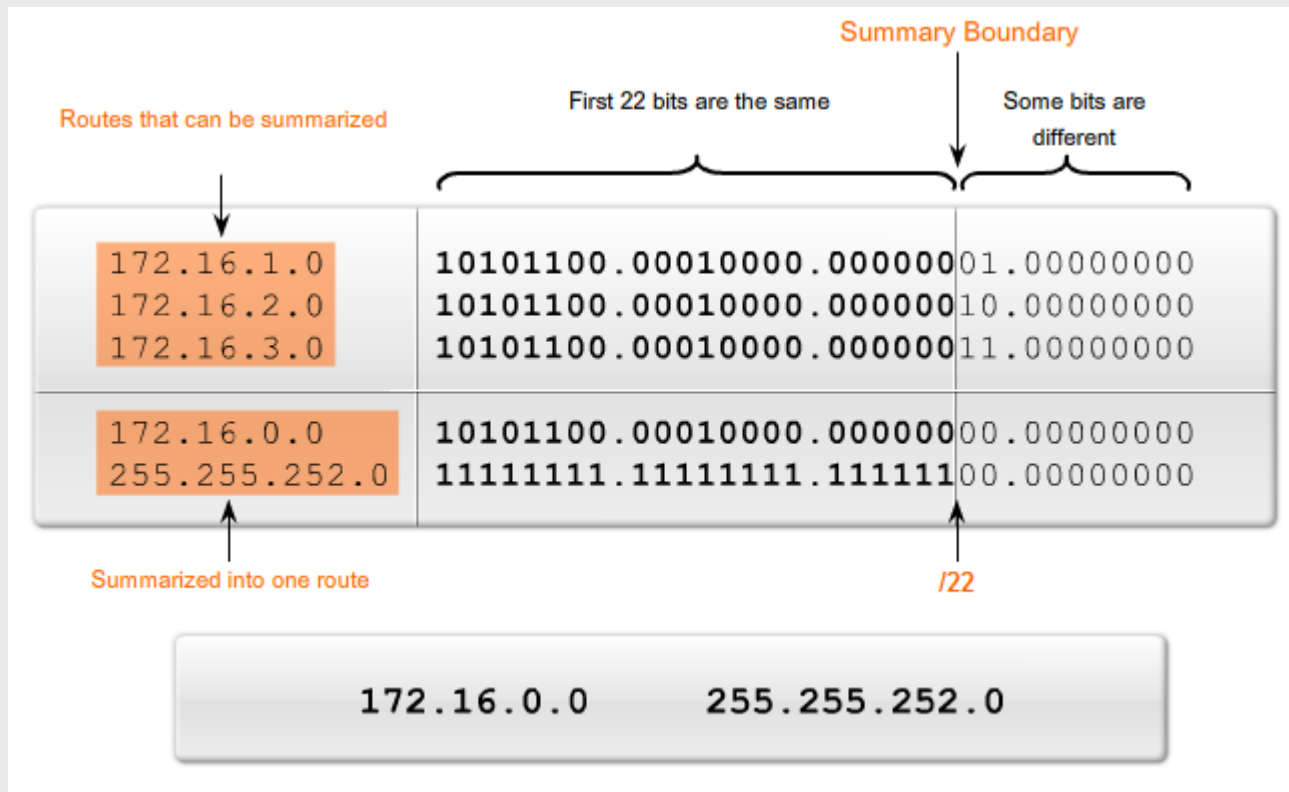
Lab activity

- The block of addresses **172.16.0.0/22** is given
 - Design an addressing scheme
 - *Design and configure static routing accordingly*

Subnet	Needed	Allocated	Address	Mask	Dec Mask	Assignable Range	Broadcast
LanC	400	510	172.16.0.0	/23	255.255.254.0	172.16.0.1 - 172.16.1.254	172.16.1.255
LanA	110	126	172.16.2.0	/25	255.255.255.128	172.16.2.1 - 172.16.2.126	172.16.2.127
LanD	90	126	172.16.2.128	/25	255.255.255.128	172.16.2.129 - 172.16.2.254	172.16.2.255
LanB	20	30	172.16.3.0	/27	255.255.255.224	172.16.3.1 - 172.16.3.30	172.16.3.31
RA-RB	2	2	172.16.3.32	/30	255.255.255.252	172.16.3.33 - 172.16.3.34	172.16.3.35
RA-RC	2	2	172.16.3.36	/30	255.255.255.252	172.16.3.37 - 172.16.3.38	172.16.3.39
RB-RC	2	2	172.16.3.40	/30	255.255.255.252	172.16.3.41 - 172.16.3.42	172.16.3.43
RD-RE	2	2	172.16.3.44	/30	255.255.255.252	172.16.3.45 - 172.16.3.46	172.16.3.47
RE-RF	2	2	172.16.3.48	/30	255.255.255.252	172.16.3.49 - 172.16.3.50	172.16.3.51

Summary routes

- Route summarization is the process of combining a number of (*static*) routes into a single (*static*) route



Summary routes

■ Configuring static summary routes

```
R3#show ip route  
<output omitted>
```

Gateway of last resort is not set

172.16.0.0/24 is subnetted, 3 subnets

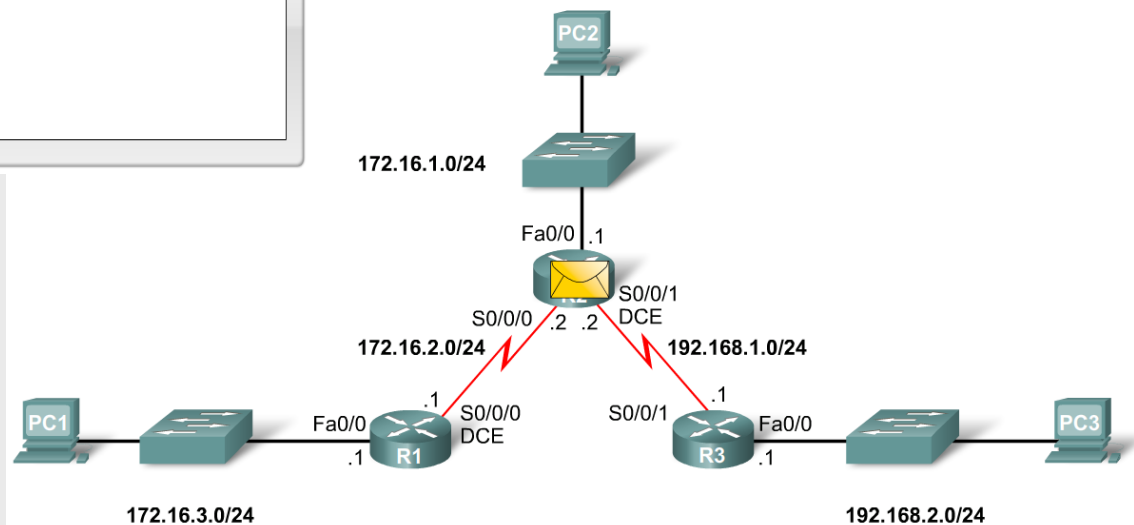
```
S 172.16.1.0 is directly connected, Serial0/0/1  
S 172.16.2.0 is directly connected, Serial0/0/1  
S 172.16.3.0 is directly connected, Serial0/0/1  
C 192.168.1.0/24 is directly connected, Serial0/0/1  
C 192.168.2.0/24 is directly connected, FastEthernet0/0
```

```
R3#show ip route  
<output omitted>
```

Gateway of last resort is not set

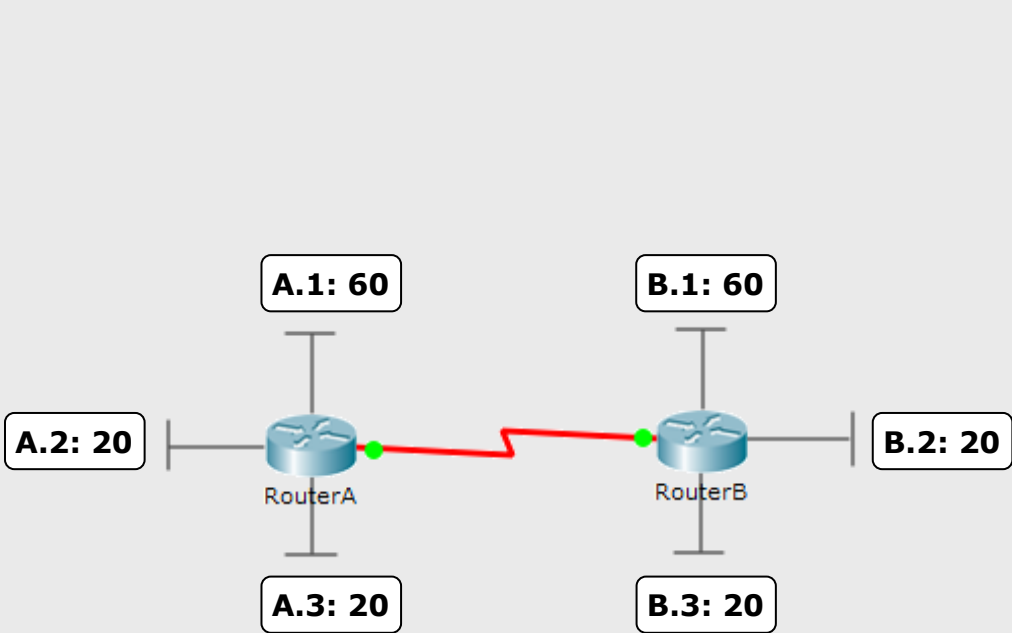
172.16.0.0/22 is subnetted, 1 subnets

```
S 172.16.0.0 is directly connected, Serial0/0/1  
C 192.168.1.0/24 is directly connected, Serial0/0/1  
C 192.168.2.0/24 is directly connected, FastEthernet0/0
```



Address planning & summarization

- Allocating prefixes by decreasing network size might be sub-optimal for the purpose of summarization



Req. addr.	Network prefix
A.1 (60)	192.168.0.0/26
B.1 (60)	192.168.0.64/26
A.2 (20)	192.168.0.128/27
A.3 (20)	192.168.0.160/27
B.2 (20)	192.168.0.192/27
B.3 (20)	192.168.0.224/27

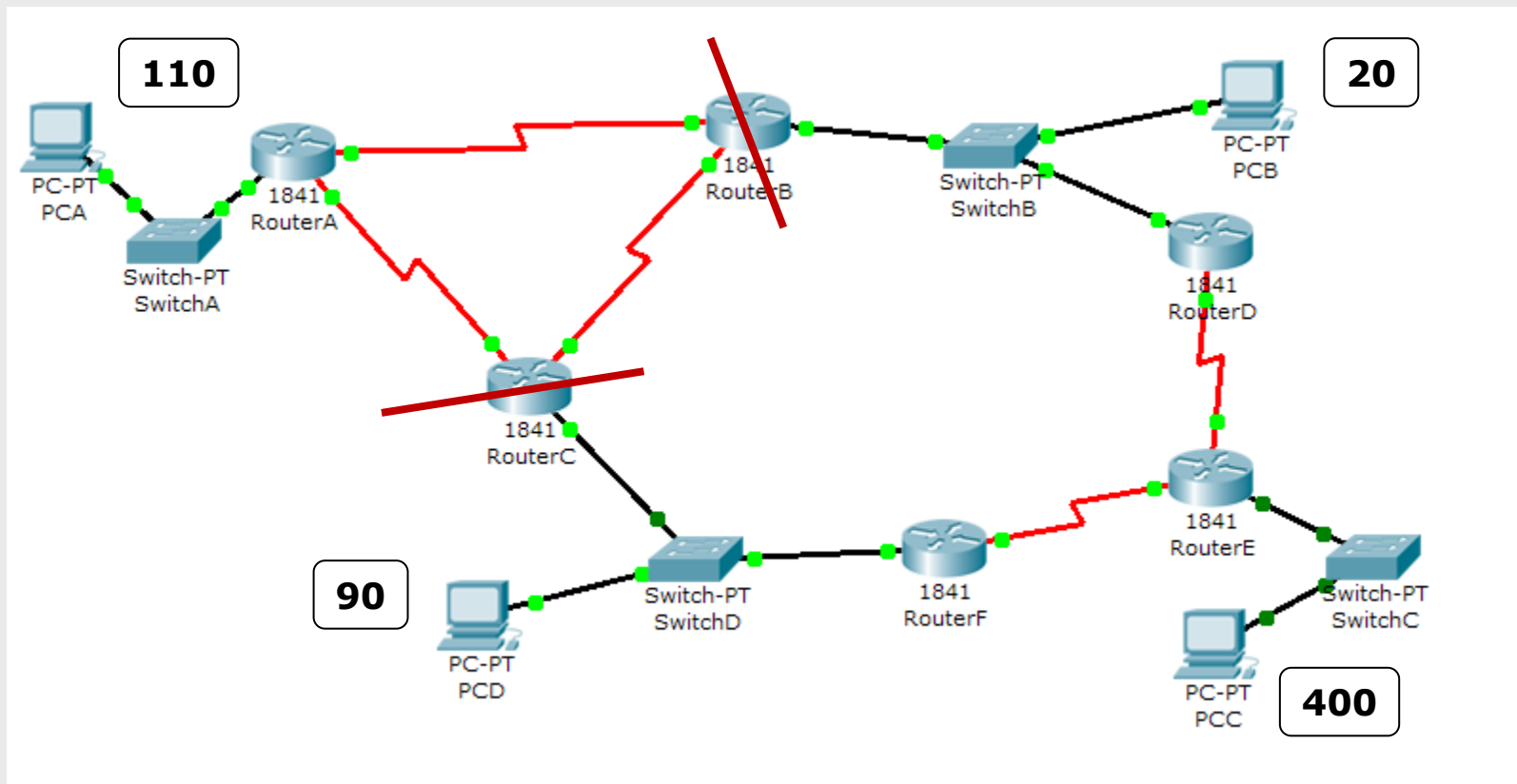
192.168.0.0/25

192.168.0.128/25

Req. addr.	Network prefix
A.1 (60)	192.168.0.0/26
A.2 (20)	192.168.0.64/27
A.3 (20)	192.168.0.96/27
B.1 (60)	192.168.0.128/26
B.2 (20)	192.168.0.192/27
B.3 (20)	192.168.0.224/27

Lab activity

- The block of addresses **172.16.0.0/??** is given
 - Design an addressing scheme to favor route summarization



Lab activity

- The block of addresses **172.16.0.0/21** is given

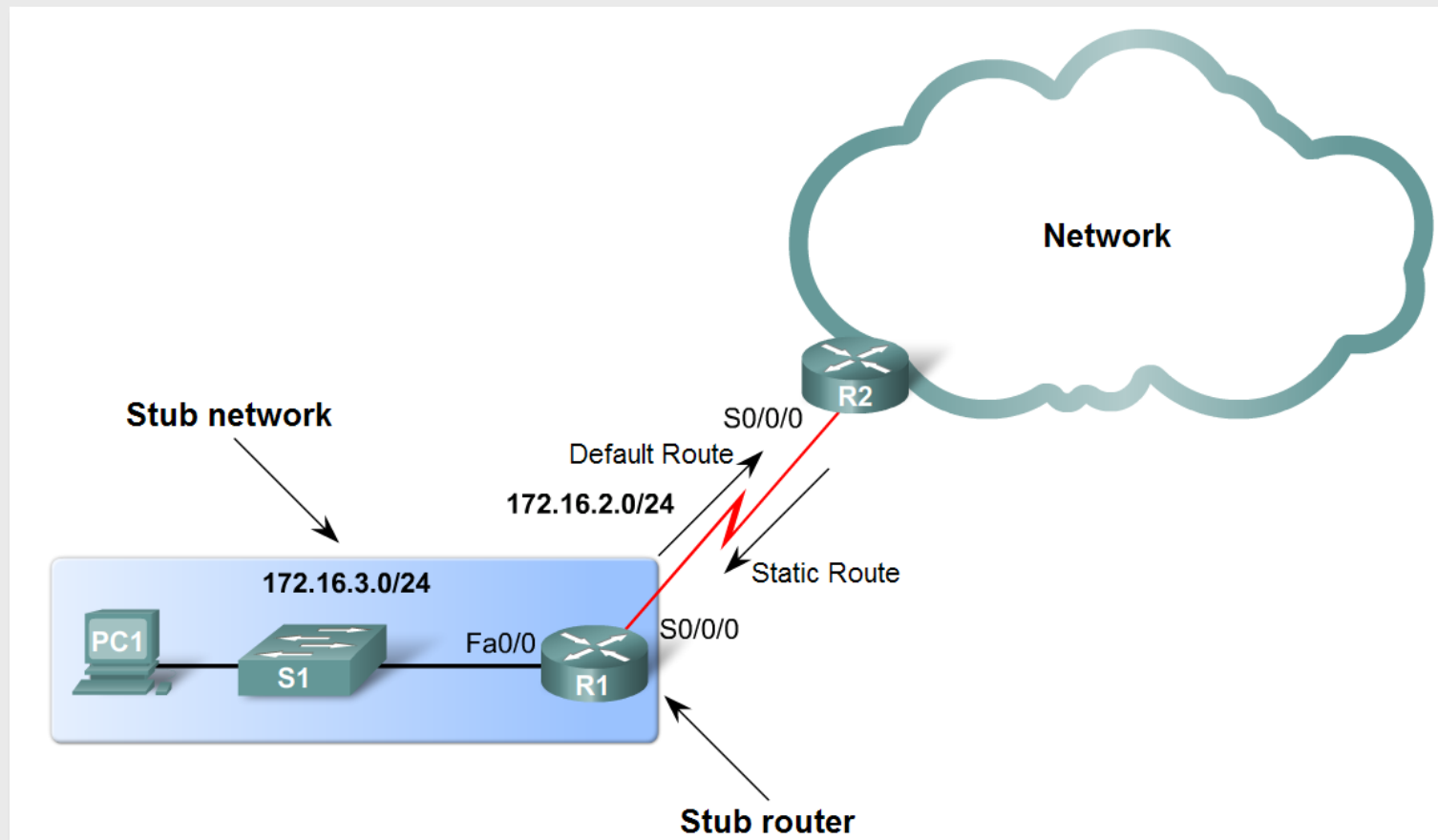
172.16.0.0/22

Subnet	Needed	Allocated	Address	Mask	Dec Mask	Assignable Range	Broadcast
LanC	400	510	172.16.0.0	/23	255.255.254.0	172.16.0.1 - 172.16.1.254	172.16.1.255
LanD	90	126	172.16.2.0	/25	255.255.255.128	172.16.2.1 - 172.16.2.126	172.16.2.127
LanB	20	30	172.16.2.128	/27	255.255.255.224	172.16.2.129 - 172.16.2.158	172.16.2.159
RD-RE	2	2	172.16.2.160	/30	255.255.255.252	172.16.2.161 - 172.16.2.162	172.16.2.163
RE-RF	2	2	172.16.2.164	/30	255.255.255.252	172.16.2.165 - 172.16.2.166	172.16.2.167
LanA	110	126	172.16.4.0	/25	255.255.255.128	172.16.4.1 - 172.16.4.126	172.16.4.127
RA-RB	2	2	172.16.4.128	/30	255.255.255.252	172.16.4.129 - 172.16.4.130	172.16.4.131
RA-RC	2	2	172.16.4.132	/30	255.255.255.252	172.16.4.133 - 172.16.4.134	172.16.4.135
RB-RC	2	2	172.16.4.136	/30	255.255.255.252	172.16.4.137 - 172.16.4.138	172.16.4.139

172.16.4.0/24

Default static route

- A default static route will match all packets for which there is no other route matching the destination address in the routing table
 - Route **0.0.0.0/0** is the default static route



Default static route

■ Configuring a default static route

- Router(config)#ip route 0.0.0.0 0.0.0.0 [exit-interface | ip-address]

```
R1#show ip route
```

```
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
```

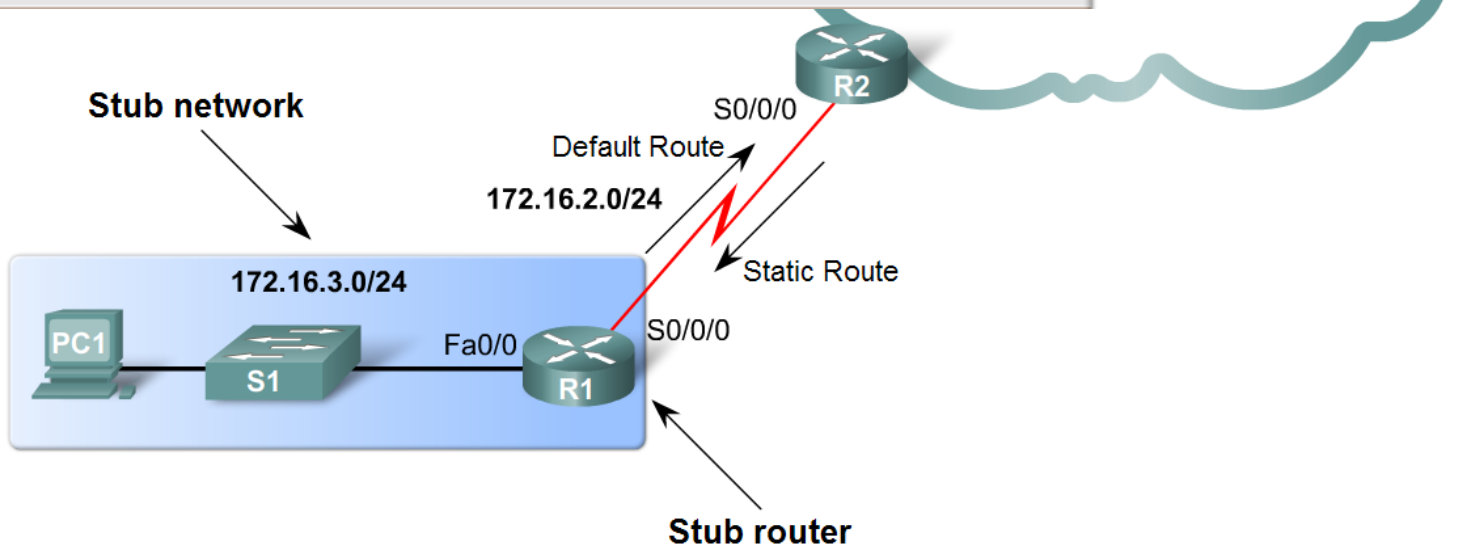
```
172.16.0.0/24 is subnetted, 2 subnets
```

```
C    172.16.2.0 is directly connected, Serial0/0/0
```

```
C    172.16.3.0 is directly connected, FastEthernet0/0
```

```
S*  0.0.0.0/0 is directly connected, Serial0/0/0
```

```
R1#
```



Troubleshooting static routes

- Connectivity troubleshooting tools
 - `ping`
 - `tracert`
 - `show ip route`
 - `show ip interface brief`
 - `show cdp neighbors detail`

